Chapter 5

Diode Applications I: Rectification

5.1 Introduction to Power Supplies

One of the most critical applications of the semiconductor diode is **Rectification**: the process of converting Alternating Current (AC) into Direct Current (DC).

Almost all agricultural electronic devices—from simple moisture sensors to complex tractor Engine Control Units (ECUs)—require a stable DC voltage source (e.g., 5V, 12V, 24V). However, power from the grid or a generator is supplied as AC (220V/50Hz).

The **DC Power Supply** performs this conversion in four distinct stages:

1. **Transformer:** Steps down high voltage AC to low voltage AC.
2. **Rectifier:** Converts AC to pulsating DC (The Diode's Job).
3. **Filter:** Smooths out the pulsations to create a steady DC.
4. **Regulator:** Maintains a fixed voltage level regardless of load changes.

In this lecture, we focus on the second block: **The Rectifier**.

Learning Objectives

By the end of this chapter, you should be able to:

- Explain the operation of Half-Wave and Full-Wave Rectifiers.
- Draw circuit diagrams and input/output waveforms.
- Calculate the DC (Average) voltage and current.
- Determine the Peak Inverse Voltage (PIV) rating required for diodes.
- Compare the efficiency and ripple frequency of different rectifier types.

5.2 Half-Wave Rectification

The simplest way to convert AC to DC is to block one half of the AC cycle. This is done using a single diode.

5.2.1 Circuit Operation

The circuit consists of an AC source (usually a transformer secondary), a single diode, and a load resistor (R_L).

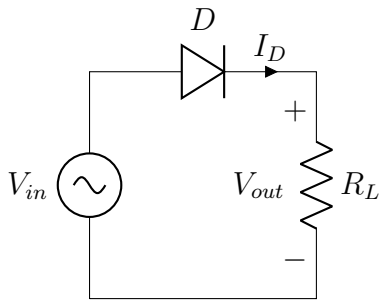


Figure 5.1: Half-Wave Rectifier Circuit.

1. **Positive Half Cycle (0 to π):**
   - The anode becomes positive with respect to the cathode.
   - The diode is **Forward Biased** (ON).
   - Current flows through the load. $V_{out} \approx V_{in}$.

2. **Negative Half Cycle (π to 2π):**

- The anode becomes negative with respect to the cathode.
- The diode is **Reverse Biased** (OFF).
- No current flows. $V_{out} = 0$.

5.2.2 Waveform Analysis

The output is a series of positive "pulses". It is unidirectional (DC) but not constant.

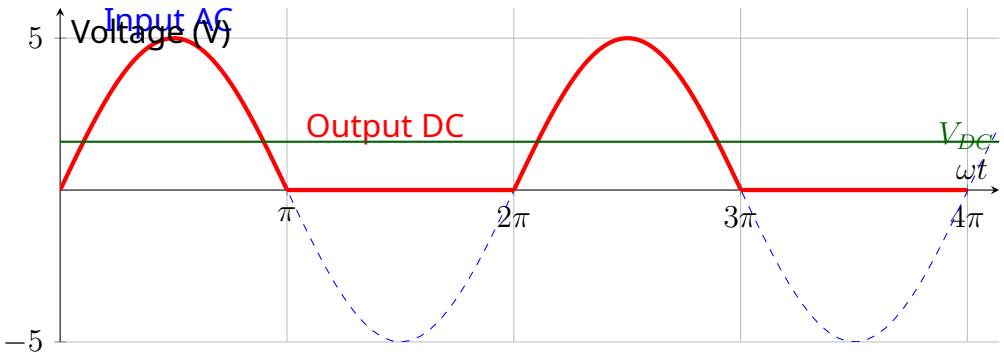


Figure 5.2: Input and Output Waveforms of a Half-Wave Rectifier.

5.2.3 Mathematical Derivations

1. Average DC Voltage (V_{dc}) The average value of a waveform is the area under the curve divided by the period (2π).

$$V_{dc} = \frac{1}{2\pi} \int_0^{\pi} V_m \sin(\theta)\, d\theta$$

Solving this integral:

$$V_{dc} = \frac{V_m}{\pi} \approx 0.318 V_m \tag{5.1}$$

Note: If using a Silicon diode, we subtract the barrier potential: $V_{dc} \approx 0.318(V_m - 0.7)$.

2. Peak Inverse Voltage (PIV) The maximum voltage the diode must withstand in reverse bias. Occurs at the peak of the negative cycle.

$$\text{PIV} = V_m \tag{5.2}$$

3. **Ripple Frequency** The output pulses occur once every cycle.

$$f_{out} = f_{in} \quad (e.g., 50 \text{ Hz}) \tag{5.3}$$

Disadvantages of Half-Wave Rectifiers

- **Low Efficiency:** It wastes the negative half-cycle. Max efficiency is only 40.6%.
- **Low DC Output:** V_{dc} is only 31.8% of peak voltage.
- **High Ripple:** The large gaps between pulses make filtering difficult.
- **Transformer Saturation:** DC current flows through the transformer core, which can cause magnetic saturation.

5.3 Full-Wave Rectification

To improve efficiency, we need to utilize *both* halves of the AC cycle. This is called Full-Wave Rectification. There are two primary methods:

1. Center-Tapped Transformer Rectifier.
2. Bridge Rectifier.

5.3.1 Center-Tapped Rectifier

This circuit uses a specialized transformer with a connection in the middle of the secondary winding (the Center Tap), effectively creating two $180°$ out-of-phase voltages.

Operation:

- **Positive Cycle:** Top of transformer is positive. D_1 is Forward Biased. D_2 is Reverse Biased. Current flows through D_1 and Load.
- **Negative Cycle:** Bottom of transformer becomes positive relative to center tap. D_2 turns ON. D_1 turns OFF. Current flows through D_2 and Load.

Crucially, in both cases, current flows through R_L in the **same direction**.

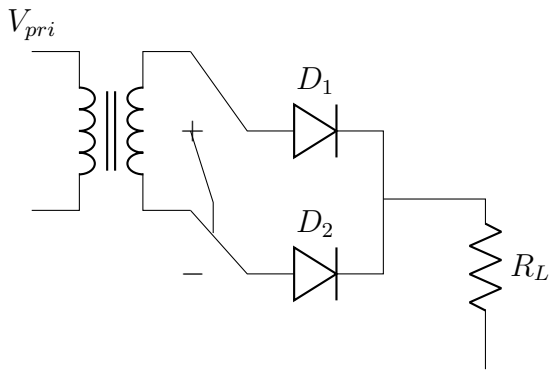


Figure 5.3: Center-Tapped Full-Wave Rectifier.

5.3.2 Analysis of Center-Tap

- **DC Voltage:** We now have two pulses per cycle.

$$V_{dc} = \frac{2V_m}{\pi} \approx 0.636V_m \tag{5.4}$$

- **PIV:** When one diode is OFF, it must withstand the peak voltage across the *entire* secondary winding ($2 \times V_m$).

$$\text{PIV} = 2V_m \tag{5.5}$$

  This is a major disadvantage. Diodes must have high breakdown ratings.

5.4 The Bridge Rectifier

This is the most popular rectifier configuration. It uses four diodes (D_1 to D_4) arranged in a diamond bridge topology. It does **not** require a bulky center-tapped transformer.

5.4.1 Operation

1. **Positive Cycle:** Input polarity is Top (+), Bottom (-).
   - Diodes D_1 and D_4 are Forward Biased.
   - Current path: Source Top $\rightarrow D_1 \rightarrow R_L \rightarrow D_4 \rightarrow$ Source Bottom.
2. **Negative Cycle:** Input polarity is Top (-), Bottom (+).

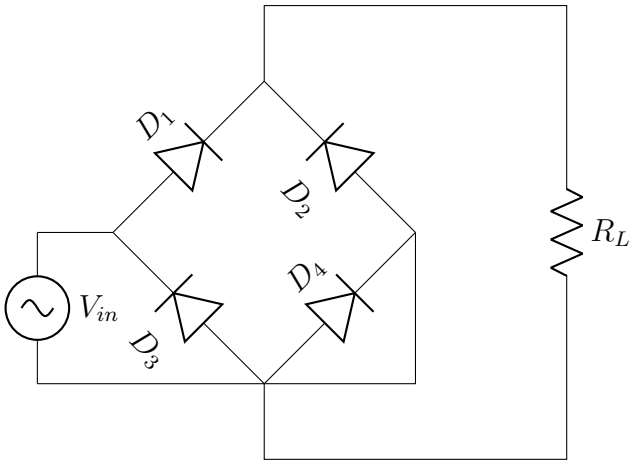


Figure 5.4: Full-Wave Bridge Rectifier Circuit.

- Diodes D_2 and D_3 are Forward Biased.
- Current path: Source Bottom $\rightarrow D_2 \rightarrow R_L \rightarrow D_3 \rightarrow$ Source Top.

5.4.2 Bridge Rectifier Characteristics

- **Efficiency:** High (81.2%).
- **DC Voltage:** Same as Center-Tap.

$$V_{dc} = 0.636 V_m \tag{5.6}$$

- **Voltage Drop:** Since two diodes are always in series, the drop is 1.4V (for Silicon).
- **PIV Rating:** Each diode only handles the peak voltage V_m.

$$\text{PIV} = V_m \tag{5.7}$$

- **Ripple Frequency:** Since there are two pulses per cycle:

$$f_{out} = 2 f_{in} \quad (e.g., 100 \text{ Hz}) \tag{5.8}$$

5.5 Comparison Summary

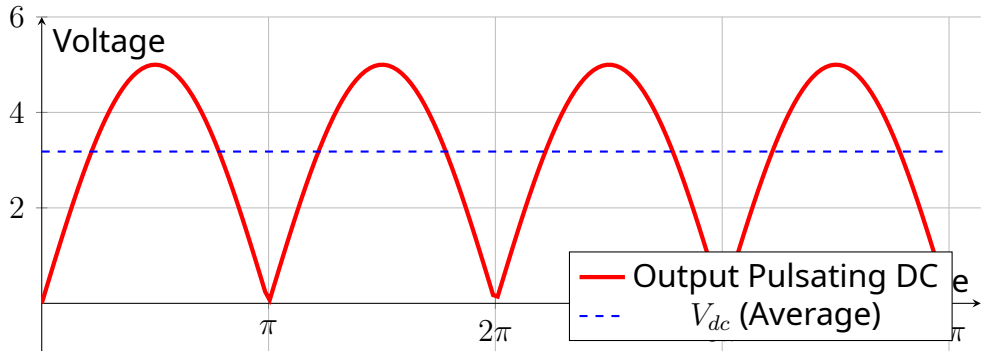


Figure 5.5: Output Waveform for Full-Wave Rectification.

Table 5.1: Comparison of Rectifier Configurations

Parameter	Half-Wave	Center-Tap FW	Bridge FW
No. of Diodes	1	2	4
Transformer Need	Normal	Center-Tapped	Normal
V_{dc} **(no load)**	$0.318V_m$	$0.636V_m$	$0.636V_m$
Ripple Factor	1.21 (High)	0.48 (Low)	0.48 (Low)
Output Frequency	f_{in}	$2f_{in}$	$2f_{in}$
PIV Rating	V_m	$2V_m$ (Costly)	V_m (Good)
Efficiency	40.6%	81.2%	81.2%

5.6 Review Questions

Part A: Conceptual Questions

1. Explain the term ”Rectification”. Why is it necessary in agricultural electronics?
2. Draw the circuit diagram of a Bridge Rectifier. Indicate the direction of current flow during the positive half-cycle.
3. Why is the output frequency of a full-wave rectifier twice the input frequency?
4. Compare Half-Wave and Full-Wave rectifiers in terms of Efficiency and Ripple Factor.
5. What is the main disadvantage of a Center-Tapped full-wave rectifier compared to a Bridge rectifier?
6. Define ”Peak Inverse Voltage” (PIV). Why is it a critical parameter when selecting a diode?

7. Why does a practical Bridge Rectifier output voltage measure slightly less than the calculated ideal value?

8. Explain why a transformer is usually included before the rectifier stage in a power supply.

9. Sketch the output waveform of a half-wave rectifier if the diode is reversed.

10. What happens to the DC output voltage of a rectifier if the input frequency is increased from 50Hz to 60Hz?

Part B: Advanced Calculations & Analysis

11. **Bridge Rectifier Analysis:** A bridge rectifier is connected to a 12V RMS (50 Hz) transformer secondary. The load resistor is 100Ω. Assuming silicon diodes (0.7V drop each):

    (a) Calculate the Peak Voltage (V_m).

    (b) Calculate the Peak Output Voltage across the load ($V_{L(peak)}$).

    (c) Calculate the DC Load Voltage (V_{dc}).

    (d) Calculate the DC Load Current (I_{dc}).

12. **Half-Wave Calculation:** A half-wave rectifier is supplied with 240V RMS. Calculate the DC output voltage assuming an ideal diode. If the diode has a PIV rating of 300V, will it survive? Explain your answer.

13. **Power Dissipation:** In the circuit from Question 11, calculate the total power dissipated by the four diodes. (Hint: Power per diode $\approx 0.7 \times I_{avg}$).

14. **Efficiency Calculation:** Prove that the maximum efficiency of a Half-Wave rectifier is 40.6%. (Start with $\eta = P_{dc}/P_{ac}$).

15. **Design Problem:** You need to design a power supply that delivers exactly 12V DC to a heater coil. You are using a Bridge Rectifier with Silicon diodes. What should be the RMS voltage rating of the transformer secondary to achieve this? (Ignore filter capacitors for this calculation).